

A review of advanced machine learning methods for the detection of biotic stress in precision crop protection

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Abstract Effective crop protection requires early and accurate detection of biotic stress. In recent years, remarkable results have been achieved in the early detection of weeds, plant diseases and insect pests in crops. These achievements are related both to the development of non-invasive, high resolution optical sensors and data analysis methods that are able to cope with the resolution, size and complexity of the signals from these sensors. Several methods of machine learning have been utilized for precision agriculture such as support vector machines and neural networks for classification (supervised learning); k-means and self-organizing maps for clustering (unsupervised learning). These methods are able to calculate both linear and non-linear models, require few statistical assumptions and adapt flexibly to a wide range of data characteristics. Successful applications include the early detection of plant diseases based on spectral features and weed detection based on shape descriptors with supervised or unsupervised learning methods. This review gives a short introduction into machine learning, analyses its potential for precision crop protection and provides an overview of instructive examples from different fields of precision agriculture.

Keywords Machine learning · Stress detection · Optical sensors · Data analysis · Plant diseases · Weed detection

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Abbreviations

FAO	Food and agriculture organization of the United Nations
EPPO	European and Mediterranean plant protection organization
SVM	Support vector machine
SVR	Support vector regression
Rbf	Radial basis function kernel
NN	Neural networks
SOM	Self-organizing maps
VI	Vegetation index
NDVI	Normalized difference vegetation index
PCA	Principal component analysis
PCs	Principal components
LDA	Linear discriminant analysis
QDA	Quadratic discriminant analysis
PLS	Partial least squares
NIR	Near infrared
RGB	Red, green and blue color image
LAB	LAB-color space: lightness (L), a and b for color-component dimensions
YCBCR	YCBCR-color space: luminance (Y), blue-yellow chrominance (CB), red-green chrominance (CR)
HSV	HSV-color space: hue, saturation, value

Introduction

Precision agriculture and precision crop protection

“Sustainable intensification”—producing more food from the same area of land while at the same time reducing the environmental impacts—demands innovative agronomic practices (FAO, Food and Agriculture Organization of the United Nations 2011). Precision agriculture strategies (the integration of different modern technologies like sensors, information and management systems) can reduce the ecological and economic impacts in agricultural crop production (Gebbers & Adamchuk 2010). According to Oerke & Dehne (2004), 40 % of the world food production is lost through weeds, diseases and insects. The occurrence of weed and disease infestation depends on specific environmental factors and exhibits a patchy distribution in the field (Franke et al. 2009). Thus the early identification of primary weed and disease foci as well as locating areas differing in weed or disease severity in the field is of high relevance (Moshou et al. 2004). Weed and pest monitoring in addition to the decision making-process are fundamental for site-specific management (Hillnhütter & Mahlein 2008). Such site-specific crop management requires a high density of spatial and temporal information.

Weeds compete with crop plants for environmental resources (such as light, water, nutrients and space) and serve as hosts for diseases and pests (Fig. 1) (Patterson 1995). Plant pathogen interactions along with the resulting disease symptoms are influenced by various external factors. Most biotic stress factors (like weeds and plant pathogens) as well as abiotic stress factors (such as water or nutrient deficiency, heat, chill or frost) reduce the yield of crops by affecting the photosynthetic apparatus (Stafford 2000), which in turn has an effect on the optical properties of plants. The observation of plants using non-invasive



Fig. 1 Crop stand of *Hordeum vulgare* and the weed *Galium aparine*. Accurate detection of *Galium aparine* is highly relevant and challenging due to a low economic threshold and high similarity to other weeds

sensors during their growth period provides new insights into the reactions of living plants to biotic stresses.

Sensor systems and challenges in data interpretation

Sensor systems, such as multi-spectral, hyperspectral and chlorophyll fluorescence, can provide high resolution data concerning agricultural crop stands (Berdugo et al. 2014; Mahlein et al. 2012a; Mahlein et al. 2013). Progress in sensor technology along with advances in computer science and geographic information systems offers new opportunities for precision agriculture, and constitutes the basis for early detection and identification of weeds and plant diseases (Chaerle et al. 2004; Lee et al. 2010; Moshou et al. 2004; Rumpf et al. 2010). The quality and quantity of data that are made available from plant sensors has dramatically increased. These sensors are highly sensitive to biological changes. However, they do not measure plant physiological parameters directly, but record a spectrum that is a sum of reflectance contributions attributed to various plant characteristics and the measurement conditions (e.g. leaf inclination, illumination and surface texture) (Jensen 2007; Gitelson et al. 2002). These observations that have high spectral, spatial and temporal resolution require the use of advanced methods of data analysis.

Traditional methods for data analysis, such as linear regression and linear discriminant analysis, are based on predefined distributions and model assumptions. These methods are applicable without loss in accuracy only for data that complies with these demands, e.g. Fisher's linear discriminant requires that the classes can be separated linearly and that each of the two classes can be represented by a unimodal Gaussian distribution (Bishop 2006). Therefore, the field of application is limited for these kinds of analysis methods.

Advanced methods of machine learning, such as k-means, support vector machines (SVMs) and neural networks (NNs) (Fig. 2), require less prior information and are applicable to a wider range of tasks as they derive the underlying distributions and model assumptions implicitly from training data (Mitchell 1997). This capacity to adapt to almost any kind of data makes them well suited for tasks with limited prior knowledge about a suitable interpretation model or complex data characteristics like non-linearity, non-Gaussian noise and outliers. These methods are particularly suitable for the interpretation of data from optical sensors because the included noise factors may be compensated by a sufficient amount of representative training data (Breiman 2001b). The resulting models are able to reduce the influence of the unknown variability significantly and provide more reliable decisions.

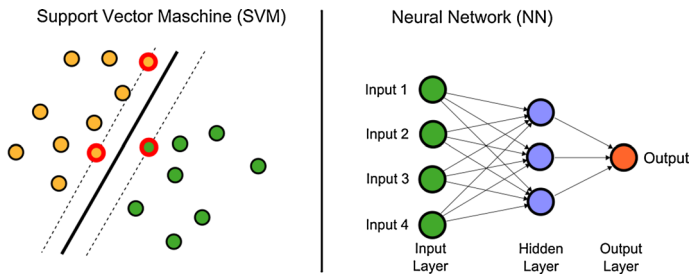


Fig. 2 Basic structures of two common machine learning methods. The linear SVM separates two classes by a discrimination function which is defined by support vectors (*red, broader circles*). The characteristics of a neural network are determined by its node structure. The example given has four input nodes in the input layer, one hidden layer with three transition nodes and one output node in the output layer (Color figure online)

A scenario with fictitious measurement values is presented in Fig. 3 and illustrates the capabilities of machine learning methods. In this scenario, a regression model of the relation between the normalized difference vegetation index (NDVI) and the severity of a disease is required. Under the assumption of limited visualization possibilities, as in real world applications, the noise distribution, the required complexity of the regression function and the number of outliers is unknown. In this scenario, the traditional method of polynomial regression is compared with a support vector regression (SVR). Figure 3 clarifies that the SVR is superior to the polynomial regression because it follows the true trend by a flexible function and is not influenced by outliers. The polynomial is not flexible enough at low orders and, at higher orders, the function starts to oscillate. Moreover, the underlying least-squares optimization is very susceptible to outliers.

Methods from the field of machine learning have been successfully applied in numerous disciplines, e.g. speech recognition (Waibel et al. 1989; Baker et al. 2009) and land cover classification in remote sensing (Friedl et al. 1999; Waske et al. 2010). In recent years, the precision crop protection community has started to benefit from the capabilities of machine learning (Karimi et al. 2006; Mucherino et al. 2009; Ahmed et al. 2012). Therefore flexible machine learning methods can help to gain new insights—and find new relationships within data for the detection of biotic and abiotic stress. This article gives an overview on common methods and reviews their applications for the detection of weeds, diseases and insects.

Machine learning for data analysis

Properties of machine learning methods

The basic assumption regarding machine learning is that computers are able to detect and quantify patterns in data by using specific algorithms (Witten and Frank 2005). This enables the machine to benefit from new information automatically—meaning the program improves by automated learning (Mitchell 1997). This allows the detection of patterns and extraction of important information from raw data even if the underlying data model is unknown (Buxton 2003; Breiman 2001b).

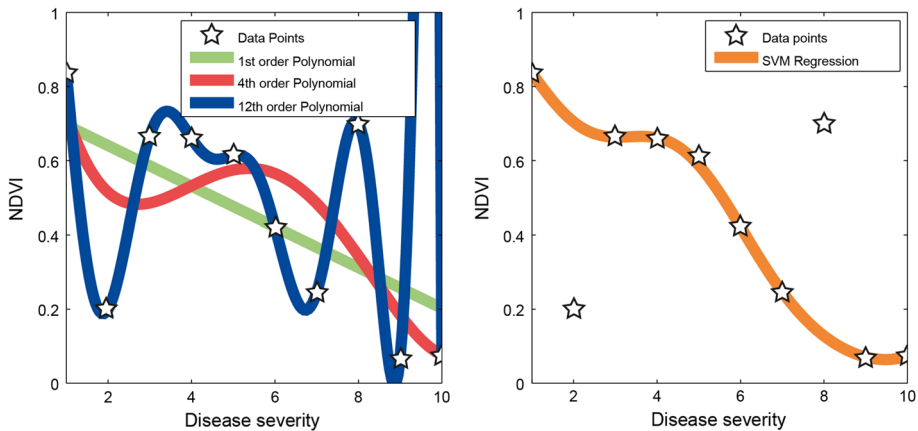


Fig. 3 Support vector regression in a regression scenario involving the NDVI Vegetation Index and the disease severity, compared to polynomial models of fixed degree and least squares optimization

Typically, the data is arranged in a matrix (X_{ij}) where each row ($x_i = [x_{i1}, x_{i2}, \dots, x_{ij}]$) is a sample composed of j features.¹ In many applications, additional target variables (named “labels” y_i) are attached to the samples (x_i). The algorithm itself derives from these training data a suitable model (F) by optimizing the model quality by a method-inherent principle. In the case of SVM classifiers, the maximum margin principle is applied (Cortes and Vapnik 1995) whereas with NNs, a non-convex optimization with regard to prediction accuracy is generally applied (Suzuki 2011). The aim of the learning process is the generation of a model that reveals task-relevant patterns without relying on unjustifiable assumptions (Buxton 2003).

There are two basic tasks for machine learning methods (Witten & Frank 2005). *Unsupervised learning* starts with unlabeled data and tries to find unknown patterns that enable a new, more compact or more comprehensive representation of the information contained (Duda et al. 1995). The most prominent example of unsupervised learning is clustering, whereby the data instances are grouped into different clusters. Normally, the cluster algorithms compose clusters which comprise samples with similar properties, as compared to the whole data set.

If target variables named *labels* y_i are given, *supervised learning* methods are applicable (Bishop 2006). Here, the identification of patterns in the data that are related to the labels (which allows for the prediction of new, unseen data) is required. A classification method is suitable if discrete label classes (e.g. *yes* or *no*) are given, and a regression method is suitable for continuous label values (Bishop 2006; Witten & Frank 2005).

The supervised learning task is divided into two phases; in phase one, a model (F) is learned by a specific machine learning algorithm from labeled training data; in phase two, it is applied to new, unknown data and predicts the best labels y for the test samples X by $y_i = F(x_i)$. In general the model quality is defined by the similarity of actual and predicted test data labels.

¹ In this article, the following notation is used: matrices are noted as bold upper-case letters in italics, vectors as bold lower-case letters in italics and scalars as lower-case letters in italics. Models and distributions which can have various representations are noted as bold upper-case letters.

In the field of precision agriculture, the labels (y_i) could be the health state of a plant (e.g. diseased or healthy) and the data samples (x_i) could be the hyperspectral reflection of a leaf surface observed by a single pixel of a hyperspectral imager (e.g. Rumpf et al. 2010).

The unsupervised and supervised learning can be implemented by different model representations (Breiman 2001b). The *generative* models include the data's joint probability distribution ($\mathbf{P}_X$) which allows for the generation of new, simulated data X_{sim} from the model (Shannon 1948).

In the case of a generative classification, an observation is usually classified by similarity measurement involving class distributions. In contrast, *discriminative* models directly describe the dependency between the data and the label values (Lafferty et al. 2001). For this purpose, modeling of the complete joint probability distribution is not needed, due to the fact that only a small part of the data is relevant for the specific task. For example, a classification task requires only information regarding the class boundary whereas the data distribution inside the classes has no effect on the classification quality. This approach of modeling only task-relevant aspects enables higher prediction accuracy and a smaller amount of training data in turn. However, this kind of model is no longer able to generate simulated samples and can give only task-related information (Breiman 2001b).

Important machine learning algorithms

The following section presents three important machine learning algorithms which have been applied in precision agriculture (Potamitis et al. 2007; Liu et al. 2010; Karimi et al. 2006).

k-Means clustering

The *k*-Means clustering algorithm is an unsupervised approach for structuring the data based on the geometrical distribution of the data points within the feature space (Bishop 2006). The observations x_j are assigned to a predefined number of clusters k which are fully defined by their central points μ_i . In an iterative process, the assignments and positions of the clusters' central points are optimized by iteratively minimizing the function.

$$J = \sum_{i=1}^k \sum_{j \in S_i} \|x_j - \mu_i\|^2, \quad (1)$$

where S_i denotes the set of all data points assigned to cluster i . This non-convex optimization results in a clustering which fits best if the data distributions support spherical clusters with similar radii. Further drawbacks are the lack of physical meaning attached to each cluster and the dependency of cluster results on the initialization (Arthur & Vassilvitskii 2007). To overcome some of the limitations, several enhancements like fuzzy k-Means (cluster probabilities; Bezdek 1981) and k-Means++ (improved initialization; Arthur & Vassilvitskii 2007) have been developed.

Neural networks

Artificial neural networks or *neural networks* are an established compact model representation for the analysis of high-dimensional data (Bishop 1995). The observations (input) and the results (output) are represented by nodes, which are inspired by the nervous system, and called neurons (Fig. 2, Rosenblatt 1962). NNs are applicable in unsupervised

and supervised contexts. The most common type of NNs is the feedforward NNs, but recurrent networks and self-organizing maps (SOM) are also used (Mitchell 1997). For modeling the relation between input and output in feedforward NN, additional neurons in so called “hidden layers” are frequently inserted between input and output neurons (Fig. 2). These nodes comprise transition functions (e.g. linear or sigmoidal functions) which transform the input signals to a local output (Bishop 1995). A network with multiple hidden layers is able to learn and represent arbitrary complex, non-linear functions (Suzuki 2011). In the learning phase of feedforward NN, the network structure, including design decisions like number of hidden layers and number of neurons per layer, has to be set and afterwards the “backpropagation” algorithm is used in most cases. This algorithm optimizes the weights of the neurons of a network with regard to a task-specific performance function (Mitchell 1997).

Due to the high number of model parameters, this algorithm is able to adapt the model to the training data with arbitrary precision (assuming enough neurons) and, therefore, can cause serious over-fitting effects. In order to counter this risk, techniques like penalty terms on the overall weight sum and the validation on a separate test data set are applied (Mitchell 1997).

The main drawbacks of applying NNs for data interpretation are the requirement to have an appropriate initial structure based on previous knowledge and that the NN is a black box with generally uninterpretable parameter values (Tu 1996).

Support vector machines

Support vector machines are used for supervised classification and regression providing linear and non-linear discrimination functions (Cortes & Vapnik 1995). SVMs contain kernel functions used to map the data requiring non-linear discrimination functions from the observation space to a potentially higher dimensional feature space. In this feature space, the data is linearly separable. The most common kernel types are the linear kernel for linear discrimination functions and the radial basis function (rbf) kernel, which enables the use of non-linear discrimination functions. The model of an SVM consists of a subset of data instances $\mathbf{x}_i$ —so called support vectors. They define a hyperplane separating classes in the feature space while its position is unambiguously defined by the principle of maximum margin, thus facilitating the learning of models with optimal generalization performance (Fig. 2). The decision function for the instance $\mathbf{x}_j$ given by a binary SVM classification model (support vectors $\mathbf{x}_i$, labels y_i , Lagrange multipliers α_i) with a kernel $\mathbf{k}(\mathbf{x}_i, \mathbf{x}_j)$ and a bias b is expressed by.

$$y_j = \text{sign} \left(\sum (y_i \alpha_i \mathbf{k}(\mathbf{x}_i, \mathbf{x}_j)) + b \right) . \quad (2)$$

The distance of a feature vector to the hyperplane can be transformed to a class probability. SVMs are easy to apply but the model representation by selected data instances makes it difficult to interpret the model.

A direct comparison of the various machine learning methods that can be applied to precision agriculture tasks is not feasible in most cases. Table 1 provides an overview of common supervised and unsupervised machine learning algorithms and characterizes them by selected categories.

The broad range of problem definitions and the “specificity” of some methods prevent meaningful comparison. For more in-depth information regarding the different methods

Table 1 Overview of unsupervised and supervised machine learning algorithms and example applications to precision crop protection tasks for the detection of weeds, plant diseases or insects (algorithms listed in alphabetic order)

Method	Learning type	Discrete/ continuous	Generative/ discriminative	Input beyond the data	Output	Assumptions	Methodical reference	Application to precision agriculture
Neural network (NN)	Supervised and unsupervised	Classification/ regression	Discriminative	Structure, functions	Predictions, probabilities	Model structure represents the problem complexity	Bishop (1995)	Liu et al. (2010): Discriminate fungal infection levels
Naïve Bayes classifier	Supervised	Classification	Generative	Distributions	Predictions, probabilities	Known class distribution, independent features	Manning et al. (2008)	Nieuwenhuizen et al. (2007): Weed detection in sugar beet fields
PLS-regression	Supervised	Regression	Discriminative	Optimization parameter	Predictions, latent variables, variable importance	Gaussian distributed data, few latent variables	Vinzi et al. (2010)	Oumar et al. (2013): Prediction of insect infestation on eucalyptus treed with hyperspectral indices
Random forest	Supervised	Classification/ feature selection	Discriminative	Number of Trees, number of tried attributes	Predictions, probabilities, feature weighting		Breiman (2001a)	Larios et al. (2010): Insect recognition
Support vector machine (SVM)	Supervised	Classification	Discriminative	C, Kernel parameter	Predictions	Kernel type corresponds to the problem	Schölkopf and Smola (2002)	Karimi et al. (2006): Stress detection in corn
Hierarchical clustering	Unsupervised	Discrete similarity information		Cluster number, distance metric	Similarity measurement, dendrogram	Non-homogeneous data distribution	Hastie et al. (2009)	Setti et al. (2009): Structuring the aggressiveness of fungal infections on peas

Table 1 continued

Method	Learning type	Discrete/continuous	Generative/discriminative	Input beyond the data	Output	Assumptions	Methodical reference	Application to precision agriculture
Image segmentation	Unsupervised	Discrete regions		Segmentation method and corresponding parameter	Segmentation in regions, partly probabilities	Reasonable regions within the image, distance metric	Dey et al. (2010)	Tang et al. (2000): Segmentation of plant and background for weed sensing
k-means	Unsupervised	Discrete cluster assignments		Cluster number k, distance metric	Centroids, cluster result	Spherical and equal sized clusters	Bishop (2006)	Potamitis et al. (2007): Acoustic identification of insects

The model type (generative/discriminative) is undefined for the unsupervised methods because they do not model a relation between data and a given label

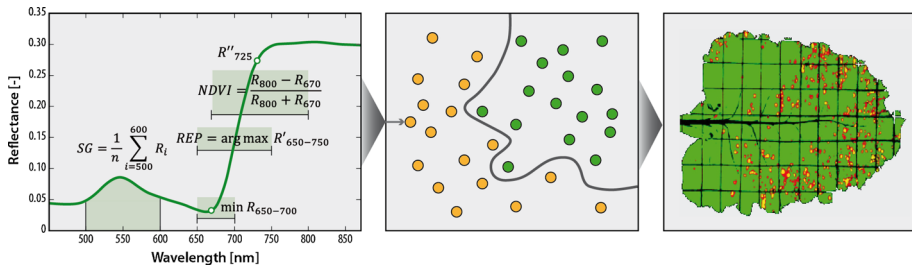


Fig. 4 The classification process, exemplified by *Cercospora beticola* infections on a sugar beet leaf. The process is divided into three phases: feature extraction from reflectance spectra, learning a classification model on training data (here a non-linear SVM) and application of the model to new data for the detection of diseased leaf pixels

available, comprehensive textbooks such as Mitchell (1997), Bishop (2006) and Witten & Frank (2005) are recommended.

Data representation and feature selection

The quality, compactness and explicitness of a learned model depend on the suitability of the data representation by features. A *feature* is a measurement of a specific property of a data sample (Guyon & Elisseeff 2003). The feature vector comprises the collection of feature values of a sample and thus describes its characteristics. From a different point of view, it describes the position of the sample in a high-dimensional feature space.

Commonly, the sensor records more values than are required for representing the relevant information. With hyperspectral measurements for stress detection, the sensor may measure the reflectance at more than one hundred wavelengths, but the relevant information can be summarized in a few features (Fig. 4; Behmann et al. 2014, Oumar et al. 2013). Reducing the number of features is an important step in data analysis as a high number of non-relevant or redundant features are able to decrease significantly the prediction accuracy—called the Hughes phenomena (Hughes 1968). Table 2 gives an overview of features from optical sensors applied to precision crop protection.

A common approach for the evaluation of hyperspectral measurements is the calculation of vegetation indices (VIs, Jensen 2007). In Fig. 4, a complete classification process is shown, including the extraction of VIs from the observed spectrum, applying a non-linear classification model, and the visualization of the predictions. The number of features can also be reduced by calculating an improved set of features from the initial set as is done by using principal component analysis (PCA, Jolliffe 2002). The PCA transforms the data into a new orthogonal feature space. With most data sets, the major part of the data variance can be expressed by the first few principal components (PCs)—resulting in the construction of a new feature set with nearly the same information. The major drawback of PCA is the focus on preservation of variance without regard to the labels and, therefore, the PCA can remove those patterns which are important for the specific task.

More specific feature selection methods use the class label to evaluate the specific relevance of features or groups of features for the specific task. As an example, the linear discriminant analysis (LDA) determines linear combination of features in consideration of the class labels and is therefore task-specific (Martínez & Kak 2001). In recent decades, several methods for feature selection have been published which can be categorized as

Table 2 Overview of features from optical sensor data for the detection of diseases, weeds and insects

Biotic stress	Sensor	Feature	Reference
Weeds	Hyperspectral	Single wavelength	Suzuki et al. 2008, Lopez-Granados et al. 2008
		Vegetation indices, waveband ratios	Lopez-Granados et al. 2008, Thorp and Tian 2004
	Fluorescence	Red fluorescence intensity (670–755 nm) and integral of the blue-green fluorescence spectra (415–455 nm)	Panneton et al. (2010), Longchamps et al. (2010)
	RGB	Principal components from fluorescence spectra Shape parameters (height, width, elongation, area to length ratio, compactness, form factor, convexity, solidity) Colour and texture features (hue, saturation, intensity, hue moments; r, g and b composition)	Arribas et al. (2011), Ahmed et al. (2012), Rumpf et al. (2012)
Plant diseases		Moment invariants (translation, scaling, rotation)	Alchanatis et al. (2005), Ahmed et al. (2012), Burks et al. (2005), Gebhardt and Kühbauch (2007)
	Hyperspectral	Single wavelengths, wavelength regions	Ahmed et al. (2012)
		Vegetation indices	Wang et al. (2008)
		Shape parameters of the curve (inclination, area under the curve)	Rumpf et al. (2010)
Fluorescence		Principal components	Laudien et al. (2003)
		Relative intensity in bi-spectral fluorescence images (550 and 690 nm)	Wu et al. (2008), Liu et al. (2010)
		Polynomials	Moshou et al. (2005)
		Photosynthesis parameters	Römer et al. (2011)
RGB		Colour and texture features (hue, saturation, grey level, contrast, entropy, variance, uniformity, mean intensity)	Bürling et al. (2010)
		Spatial information	Camargo and Smith (2009), Pydipati et al. (2005), Wetterich et al. (2013)
		Shape parameters (length, extend, major axis etc.)	Zhang et al. (2011)
		Single wavelengths	Camargo and Smith (2009)
Insects	Hyperspectral	Vegetation indices, wavelength ratios	Mirík et al. (2007)
		Shape of the insect	Mirík et al. (2007), Prabhakar et al. (2011)
	RGB	Shape of the damage	Batista et al. (2010), Larios et al. (2010)
			Zhao et al. (2012)

either filter, wrapper or embedded methods (Saeys et al. 2007). Wrappers utilize a learning algorithm and quantify the combined relevance of a feature subset by the quality of the predicted output (Kohavi & John 1997). Filter methods use inherent characteristics of features to estimate a feature ranking according to their relevance (Robnik-Šikonja & Kononenko 2003). Embedded methods rank features by using relevance indicators derived during the training phase of a specific learning algorithm (Rakotomamonjy 2003).

Choice of machine learning methods

The choice of an optimal data analysis method strongly depends on the specific problem. Therefore, it is not possible to provide general recommendations, but some criteria could help to identify applicable algorithms.

There are some basic properties that are important for a method selection: the number of features, the number of training samples, the type of label and information about the data distributions. Assuming that label information is provided, a supervised method should be used. Without labels, only unsupervised methods can be used. Discrete label values demand a classification; continuous ones demand a regression.

Experience with the data and the type of analysis result can be used to select a suitable algorithm (Jebara 2004). For tasks with strong prior knowledge about the data distribution, generative models should be used. These models can be used to answer many potential queries as the distributions of the data are included. However, in many cases these generative models are outperformed by discriminative models (Lafferty et al. 2001). Discriminative learning algorithms are more robust against wrong assumptions and do not rely on prior information to the same extent (Breiman 2001b). In cases with weak information on data distribution or required model structure, discriminative models like SVM and Random Forest (Breiman 2001a) should be applied.

Generally, all information available, like specific distributions, relations between samples and label structure should be included in the learning process. For instance, single pixels of images need to be analyzed with respect to their local neighborhood (Dey et al. 2010). If prior information about similarities of classes (e.g. about the weed species in Fig. 5) is available, this knowledge should also be introduced in the analysis—as shown in Rumpf et al. (2012). They performed a sequential classification of crop and weed species. The applied classification tree was designed based on known similarities and economic thresholds (Fig. 5).

In a few publications, more than one algorithm is applied, enabling a direct comparison of algorithms to be made—at least with regard to specific data. Karimi et al. (2006) showed that an SVM approach achieves a better accuracy than an NN approach (69 to 58 %) for the detection of the combined stress effect of weeds and nitrogen in maize. Garcia-Ruiz et al. (2013) demonstrated that a non-linear SVM (85 % accuracy) is superior to a linear SVM and linear and quadratic discriminant analysis (LDA and QDA; 75 % accuracy and less) for the detection of citrus greening disease using aerial, multi-spectral images. In most of the studies, non-linear approaches such as non-linear SVM and NN reach higher prediction accuracies than linear approaches. However, their application is accompanied by a much higher model complexity and an elevated risk of over-fitting. A generalization of the prediction accuracies is not justified as the data sets and also the configurations of the algorithms are very specific.

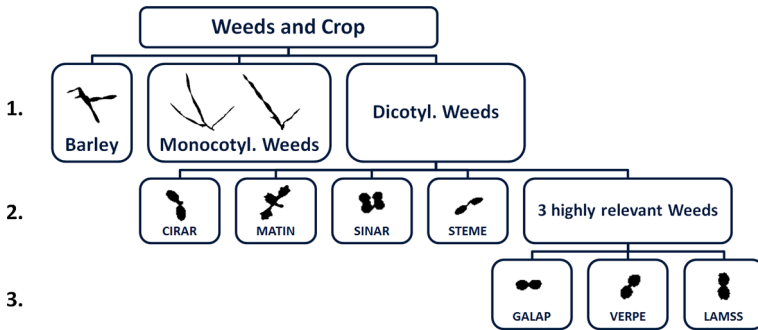


Fig. 5 Decision tree of the sequential classification of nine weeds and the crop *Hordeum vulgare* at an early growth stage according to Rumpf et al. (2012). In the first classification step, a differentiation between the three groups, crop, monocotyledonous and dicotyledonous weeds was undertaken. Subsequently, specific discrimination within the group of dicotyledonous weeds was undertaken and, following that, the challenging discrimination between highly relevant weeds with similar phenological appearance was undertaken. The species are identified by their EPPO-Codes (EPPO, European and Mediterranean Plant Protection Organization 2014)

Application of machine learning for precision crop protection

In precision agriculture, machine learning methods and pattern recognition are still at an early stage of development (Mucherino et al. 2009; Huang et al. 2010). Within the last decade, these methods have also become of interest in precision crop protection and plant sciences for the early detection and identification of biotic stress (Moshou et al. 2004; Rumpf et al. 2010; Zhang & Meng 2011). The main aim of using machine learning methods in precision crop protection is to detect variability or heterogeneity within crop stands caused by biotic stresses (such as diseases or weeds) from sensor data. Remote and near-sensed optical data has been widely explored as a possible data source for the detection and mapping of weeds and plant diseases in agricultural crops (Lamb & Brown 2001; Mahlein et al. 2012b; Mewes et al. 2011; Thorp & Tian 2004). It turns out that machine learning methods can improve both the speed and accuracy of data analysis, compared to traditional statistical discrimination models. Below, a brief literature overview on the application of machine learning methods for the detection of weeds, plant diseases and insects is given.

Detection of weeds

Accurate weed identification in crop stands is a prerequisite for applying control strategies, such as site-specific herbicide application or mechanical methods. A range of learning approaches has been developed for weed identification. Most of them utilize morphological properties and shape parameters of weed leaves as features for subsequent classification (Table 2). In addition, information from reflectance spectra is used to identify characteristic patterns of weed plants. It has been shown in recent years that using discriminative features is crucial for weed detection from sensor data. Image segmentation and digitizing combined with subsequent classification are promising approaches for the identification of crop plants and weeds at different stages during the vegetative period.

Hyperspectral imaging and non-imaging data

Karimi et al. (2006) used SVMs and NNs for weed and nitrogen deficiency detection in corn with non-imaging hyperspectral reflectance data as input. In an extensive approach, they uniquely identified and classified four weed management practices and three different nitrogen rates. The classification accuracy using SVMs was higher than with NN in this research project—69.2 versus 58.3 % respectively. Lopez-Granados et al. (2008) used the hyperspectral reflectance spectrum from 400 to 900 nm for the classification of late season grass weeds and wheat plants in a field study. Their approach using linear and non-parametric functional discriminant analysis and NNs has shown that, in general, a preliminary computation of most relevant PCs improves the classification accuracy. They concluded from their study that analysis in real time of high spectrally resolved images will be adequate to map grass weed patches in wheat. However, it has to be mentioned that the differentiation between grass weeds and cereal plants is challenging due to their phenological similarity as both are monocotyledonous plants. Segmentation of hyperspectral imagery by explanatory variables using LDA and NNs was demonstrated by Suzuki et al. (2008). First, NDVI was calculated with hyperspectral images to differentiate between plants and background soil using a threshold value. For discrimination between soybean crop plants and weed plants, 15 relevant wavebands as explanatory variables were selected using a filter approach. Most relevant wavelengths were located around the green peak and in the near infrared (NIR) range. For discrimination purposes, a subset of the relevant features was determined using a filter or PCA. These methods were combined with LDA or NN and evaluated regarding accuracy and efficiency (speed). In terms of accuracy, NN models were superior to LDA models; PCA models were superior to the filter. Models using a filter however, used less computation time than PCA models. For practical implementation, Moshou et al. (2002) developed a weed species spectral detector based on neural networks with local linear mappings. A self-organized map (SOM) neural network achieved fast convergence and good generalization. The proposed method classified crops and different species of weed with high accuracy.

Chlorophyll fluorescence

Longchamps et al. (2010) and Panneton et al. (2010) used the ultraviolet-induced fluorescence of plants as an alternative for weed detection in corn. Based on a partial least squares (PLS) model for classification, two large bands were chosen in the blue-green fluorescence zone (400–425 nm and 425–490 nm). A linear discriminant analysis using the signal from these two bands provided a robust differentiation between monocotyledonous and dicotyledonous plants. It was possible to classify corn hybrids and dicotyledonous weed species with a success rate of 91.8 %. However, the separate identification of corn and monocotyledonous weed species was difficult even under the tightly controlled conditions of the experiment. Thus the application to field conditions requires further research.

RGB-images

RGB- color images (RGB) with the three channels red, green and blue are used in different studies for the detection of biotic stress. Color, shape and texture analysis methods have been shown to be successful for the detection of weeds in crop stands (Perez et al. 2000). In both cases, a systematic selection of relevant features from the RGB-images increased species classification accuracies—as indicated by Burks et al. (2005). A NN classifier with

texture features achieved excellent weed species classification accuracies of 96.7 % and high individual species classification accuracies above 90.0 % (Burks et al. 2005). In their study (under controlled conditions), reduced variable data sets performed better than the full unreduced data set. They stated that NN classifiers are well suited for real-time control applications and that vision systems can achieve real-time discrimination of both weed species and weed infestation levels to provide information for management strategies. Texture features can also be extracted from gray level images as described by Wu & Wen (2009). The space dimension was reduced by PCA, resulting in higher classification accuracies from 92.3 to 100 %—comparable to results obtained by Lopez-Granados et al. (2008). Additionally, the best classification performance was achieved in their study by SVMs compared with NNs. Arribas et al. (2011) proposed pattern recognition based on four main stages: After data acquisition, a first segmentation based on the RGB color space was performed. From these segmented images, relevant shape features such as the major and minor ellipse axes or the area were extracted and a set of most discriminative features was selected. Classification was performed using neural networks, classifying sunflower from weed leaves with an accuracy of 85 %. For an automated weed classification system, Ahmed et al. (2012) defined 14 features to characterize crops and weeds by SVMs from digital images, with an accuracy of 97 %. As the economic losses due to weeds differ according to species, and different weeds are controlled by different herbicides, correct identification of the species is required. Thus Rumpf et al. (2012) developed a sequential SVM classification for small-grain weed species by shape parameters from image data (Fig. 5). Here the difficulty was to distinguish especially within phenologically similar weed species such as *Veronica persica* and *Galium aparine*. The authors firstly classified similar sub-groups such as crop plants, monocotyledonous weeds and dicotyledonous weeds. In a second step, weed species within one group were differentiated. Specialized features for each sub-group were selected by the embedded approach of SVM-weighting and the filter approach RELIEF-F (Robnik-Šikonja & Kononenko 2003). For the classification task, both linear and non-linear SVMs were used. Using this approach, an overall classification accuracy of 97.7 % was achieved. This approach benefits by a reduction of the complexity of the classification problem by dividing it into individual, less complex parts. Different automatic leaf image classification systems have been developed for selective herbicide application (Gerhards & Sökefeld 2003; Timmermann et al. 2003; Yang et al. 2003). Here, the classification result is directly linked to spatial information and herbicide application maps are created. Results from a field trial with real time image analysis and GNSS-controlled patch spraying showed high efficacy in weed control by 85–98 %; and the amount of herbicides applied was reduced by between 6–81 % (Gerhards & Oebel 2006). In general, one can conclude that for weed detection and classification, imaging-based approaches are superior to non-imaging approaches. These studies underscore the benefit of machine learning and inter-disciplinary approaches for precision crop protection in practice.

Detection of plant diseases

Detecting and forecasting plant diseases in the field provides a basis for sustainable and effective plant protection. In this context, machine learning is applied to different kinds of sensor data.

Hyperspectral imaging and non-imaging data

One of the first approaches to differentiate biotic stress of plants using machine learning was realized by Moshou et al. (2004). They automatically detected yellow rust in wheat on the basis of reflectance measurements using NNs. Using a total of 5137 leaf spectra for evaluation, a classification performance between 95 and 99 % was obtained. Then Moshou et al. (2005) defined specifications for the development of a cost-effective optical device for field measurements. This system simultaneously obtains hyperspectral, multi-spectral images and GNSS information of crop stands. In an intelligent multi-sensor fusion decision system based on NNs, the presence of diseases can be predicted in the field (Moshou et al. 2011). Wang et al. (2008) were able to predict *Phytophthora infestans* infections on tomato plants by relevant regions of hyperspectral signatures using NNs. They identified diseased tomato plants when first symptoms were visible. Wu et al. (2008) showed that early detection of *Botrytis cinerea* on eggplant leaves is possible by applying PCA before NN classification using hyperspectral signatures as features. In their study, first small symptoms of grey mold on eggplant leaves were accurately detectable. Liu et al. (2010) detected rice glume blight disease in rice panicles combining PCA and subsequent NN classification, differentiating between healthy, light, moderate and serious infection levels. However, this study was performed under controlled conditions and they stated that it is probably not applicable for monitoring the health status of rice under field conditions. Rumpf et al. (2010) published one of the first studies demonstrating early detection of plant diseases before visible symptoms appeared by using SVMs. Nine spectral vegetation indices, related to physiological parameters, were used as features for automatic classification of relevant sugar beet diseases. Early differentiation between healthy and inoculated plants as well as among specific diseases (*Cercospora* leaf spot, leaf rust and powdery mildew) was achieved by a non-linear Support Vector Machine with high accuracy (up to 90 %). However further research is required to bring the described methods into field application for precision crop protection.

Chlorophyll fluorescence

A few research groups have applied machine learning methods to chlorophyll fluorescence data. Chaerle et al. (2004) have shown that non-destructive chlorophyll fluorescence techniques were able to detect infections by plant diseases at an early stage. Römer et al. (2011) thus applied a polynomial approximation of fluorescence curves for feature extraction before classification with SVMs. Polynomials of low order were used for a piecewise approximation of the whole fluorescence curve of wheat leaves inoculated with *Puccinia triticina*. The polynomial coefficients contain the information about the form of the approximated piece of the curve. It turned out that wavelengths between 450 and 500 nm and between 550 and 630 nm were able to separate healthy leaves from leaves inoculated with *P. triticina*. A classification accuracy of 93 % was achieved on the second day after inoculation. At this early stage of infection, no symptoms of wheat leaf rust were visible. A recent publication by Wetterich et al. (2013) classified Huanglongbing-infection (citrus greening disease) of citrus trees from fluorescence imaging spectroscopy. After texture features, such as autocorrelation, contrast and energy were extracted, a SVM classification was performed. The authors were able to apply and transfer this approach to two different chlorophyll fluorescence data sets from Brazil and the USA with classification accuracies above 90 %.

RGB-images

Several research groups have used pattern recognition methods and machine learning to detect and to identify plant diseases from RGB-images. Along with color information in the RGB, LAB, YCBCR or HSV color space, the spatial information provides important characteristics of plant diseases (Bock et al. 2010). Single pixels can be analyzed with respect to their local neighborhood (Dey et al. 2010). Shape parameters and parameters correlated to the appearance of the objects such as grey levels, texture, dispersion and connectivity were defined as features for the detection and identification of disease symptoms in plants by Camargo & Smith (2009). The proposed features were used for SVM classification. Since not all features are supposed to give the same amount of information on characteristic disease symptoms, a cross-validation helped to identify an optimized feature sub-set for classification. Symptoms are not uniform areas; in fact, they are complex regions composed of different cellular situations such as chlorotic or necrotic areas, water-soaked margins or dry and dead lesions. These different patterns can be used as characteristics for identification by image analysis methods as presented by Zhang & Meng (2011). They used a hierarchical approach to detect and differentiate citrus canker, a bacterial disease of citrus tree leaves, from other damage and disease symptoms in the field.

In summary, image analysis, chlorophyll fluorescence and hyperspectral techniques combined with machine learning techniques are powerful approaches for early detection of biotic stress. However, the diagnostic potential based on imaging and chlorophyll fluorescence data alone is limited. Spectral approaches are more promising for this task. A deeper understanding of hyperspectral signatures of diseased plants is needed to model how several input variables affect the resulting spectra and how these variables are correlated. Machine learning, especially feature selection methods combined with classification methods, could cope with this complex challenge.

Detection of insects

The detection of insects and pests by optical sensors and machine learning is the focus of only a few research groups. In this research field, the calculation of VIs from spectral data and the correlation to damage intensity is more common. Mirik et al. (2007) assessed the relationship between VIs and Russian wheat aphid abundance in wheat canopies. In this study, an aphid index based on reflectance at 576, 908, 756 and 716 nm showed consistent relationships with aphid occurrence in the field. A specific spectral index to detect and to quantify the infestation of cotton with leafhopper was introduced by Prabhakar et al. (2011). Relevant wavelengths were identified by plotting linear intensity curves between reflectance and the infestation state. This resulted in six sensitive wavebands which were combined into ratio indices. Two indices with the best performance were selected by regression analysis and were used to detect gradients of leafhopper infestation in cotton. Zhao et al. (2012) used hyperspectral imaging data to quantify infestation of oilseed rape by cabbage caterpillar. They were able to count the percentage of wormhole areas on a leaf by a hole filling function. To assess wormholes on leaf edges, an adapted NN reconstruction algorithm was proposed. This approach was superior to other neural network algorithms. In terms of insect species identification, some research was undertaken based on RGB data. Moths were correctly identified by Batista et al. (2010) using a simple approach based on the k-nearest neighbor classifier, combining distance measures for texture, color and shape. Larios et al. (2010) proposed an image classification method for

the detection of stonefly larvae. Complex texture features combined with SVMs were able to differentiate between different insect species and to identify stonefly larvae correctly.

However, one of the difficulties in detecting insect pests with optical sensors is that living insects are not easy to photograph or to measure. For precision crop protection applications, it is more practicable to assess the damage at an earlier stage than detecting the insects themselves.

Conclusions

In the last 25 years, the sensing technologies used in precision agriculture have advanced substantially producing large amounts of data from crop stands. Machine learning methods are able to analyze high dimensional data with unknown statistical characteristics for precision crop protection by learning the model structure directly from training data. The discussion of machine learning focuses on different algorithms but the process of feature selection is important. Convincing analysis approaches combine a suitable feature extraction and selection with appropriate prediction algorithms. We are at the beginning of a promising development which is highly relevant to the improvement of crop yields. However the potential is far from being exploited. Based on the current research dynamics in sensing technologies and methods for data analysis, the following trends for the future of data analysis in precision crop protection are expected:

1. Further adaptation of established machine learning methods to plant specific tasks. A possible focus will be the combined analysis of spatial and spectral data dimensions, additionally incorporating temporal data.
2. Faster, higher resolution and more accurate sensor systems with integrated data analysis modules based on machine learning methods becoming available for mobile applications. These sensor systems will support automated decision making processes and adequate visualization of results directly in the field.
3. The combination of data from different sensor types for comprehensive plant models. In combination with a time-dependent component, these task-independent models will be able to predict the future states within a field—and take the current weather forecast into consideration at the same time.

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References

- Ahmed, F., Al-Mamun, H. A., Bari, H. A. S. M., Hossain, E., & Kwan, P. (2012). Classification of crops and weeds from digital images: a support vector machine approach. *Crop Protection*, 40, 98–104.

- Alchanatis, V., Ridel, L., Hetzroni, A., & Yaroslavsky, L. (2005). Weed detection in multi-spectral images of cotton fields. *Computers and Electronics in Agriculture*, 47(3), 243–260.
- Arribas, J. I., Sanches-Ferrero, G. V., Ruiz-Ruiz, G., & Gomez-Gil, J. (2011). Leaf classification in sunflower crops by computer vision and neural networks. *Computers and Electronics in Agriculture*, 78(1), 9–18.
- Arthur, D., & Vassilvitskii, S. (2007). k-means++: the advantages of careful seeding. In: N. Bansal, K. Pruhs, C. Stein (Eds.), *Proceedings of the eighteenth annual ACM-SIAM symposium on discrete algorithms (SODA)* (pp. 1027–1035). Society for Industrial and Applied Mathematics, Philadelphia, PA, USA.
- Baker, J., Deng, L., Glass, J., Khudanpur, S., Lee, C. H., Morgan, N., et al. (2009). Developments and directions in speech recognition and understanding, Part 1. *IEEE Signal Processing Magazine*, 26(3), 75–80.
- Batista, G. E. A. P. A., Campara, B., Keogh, E. (2010). Classification of live moths combining texture, color, and shape primitives. In: S. Draghici, T. M. Khoshgoftaar, V. Palade, W. Pedrycz, M. A. Wani X.Zhu (Eds.), *Proceedings of the Ninth International Conference on Machine learning and Application*, IEEE, (pp 903–906). New York, NY, USA.
- Behmann, J., Steinrücken, J., & Plümer, L. (2014). Detection of early plant stress responses in hyperspectral images. *ISPRS Journal of Photogrammetry and Remote Sensing*, 93, 98–111.
- Berdugo, C., Zito, R., Paulus, S., & Mahlein, A.-K. (2014). Fusion of sensor data for the detection and differentiation of plant diseases in cucumber. *Plant Pathology*,. doi:10.1111/ppa.12219.
- Bezdek, J.C. (1981). *Pattern recognition with fuzzy objective function algorithms*, Series of Advanced applications in pattern recognition. Plenum Press, New York, USA.
- Bishop, C. M. (1995). *Neural networks for pattern recognition*. Oxford, UK: Oxford University Press.
- Bishop, C. M. (2006). *Pattern recognition and machine learning*. New York, USA: Springer.
- Bock, C. H., Poole, G. H., Parker, P. E., & Gottwald, T. R. (2010). Plant disease severity estimated visually, by digital photography and image analysis, and by hyperspectral imaging. *Critical Reviews in Plant Science*, 29(2), 59–107.
- Breiman, L. (2001a). Random forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L. (2001b). Statistical modeling: the two cultures. *Statistical Science*, 16(3), 199–231.
- Burks, T. F., Shearer, S. A., Heath, J. R., & Donohue, K. D. (2005). Evaluation of neural-network classifiers for weed species discrimination. *Biosystems Engineering*, 91(3), 293–304.
- Buxton, H. (2003). Learning and understanding dynamics scene activity: a review. *Image and Vision Computing*, 21(1), 125–136.
- Camargo, A., & Smith, J. S. (2009). Image pattern classification for the identification of disease causing agents in plants. *Computers and Electronics in Agriculture*, 66(2), 121–125.
- Chaele, L., Hagenbeck, D., De Bruyne, E., Valcke, R., & Van der Straeten, D. (2004). Thermal and chlorophyll-fluorescence imaging distinguish plant-pathogen interactions at an early stage. *Plant Cell Physiology*, 45(7), 887–896.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine learning*, 20(3), 273–297.
- Dey, V., Zhang, Y., Zhong, M. (2010). A review on image segmentation techniques with remote sensing perspective. In: W. Wagner, B. Székely (Eds.), *Proceedings of the International Society for Photogrammetry and Remote Sensing (ISPRS) TC VII Symposium, IAPRS*, 38(7A), (pp. 5–7).
- Duda, R. O., Hart, P. E., & Stork, D. G. (1995). *Pattern Classification and Scene Analysis* (2nd ed.). Hoboken, NJ, USA: John Wiley & Sons Inc.
- EPPO, European and Mediterranean Plant Protection Organization (2014). EPPO plant protection thesaurus, (<http://eppt.eppo.org/#>).
- FAO, Food and Agriculture Organization of the United Nations. (2011). *Save and grow*. Rome, Italy: FAO, Food and Agriculture Organization of the United Nations.
- Franke, J., Gebhardt, S., Menz, G., & Helfrich, H.-P. (2009). Geostatistical analysis of the spatiotemporal dynamics of powdery mildew and leaf rust in wheat. *Phytopathology*, 99(8), 974–984.
- Friedl, M. A., Brodley, C. E., & Strahler, A. H. (1999). Maximizing land cover classification accuracies produced by decision trees at continental to global scales. *IEEE Transactions on Geoscience and Remote Sensing*, 37(2), 969–977.
- Garcia-Ruiz, F., Sankaran, S., Maja, J. M., Lee, W. S., Rasmussen, J., & Ehsani, R. (2013). Comparison of two aerial imaging platforms for identification of Huanglongbing-infected citrus trees. *Computers and Electronics in Agriculture*, 91, 106–115.
- Gebbers, R., & Adamchuk, V. I. (2010). Precision agriculture and food security. *Science*, 327(5967), 828–831.

- Gebhardt, S., & Kühbauch, W. (2007). A new algorithm for automatic *Rumex obtusifolius* detection in digital images using colour and texture features and the influence of image resolution. *Precision Agriculture*, 8(1–2), 1–13.
- Gerhards, R., & Oebel, H. (2006). Practical experiences with a system for site-specific weed control in arable crops using real-time image analysis and GPS-controlled patch spraying. *Weed Research*, 46(3), 185–193.
- Gerhards, R., & Sökefeld, M. (2003). Precision farming in weed control—system components and economic benefits. In: J. V. Stafford, A. Werner (Eds.), *Precision Agriculture '03 Proceedings of the 3rd European Conference on Precision Agriculture*, (pp 229–240). Wageningen Academic Publishers, The Netherlands.
- Gitelson, A. A., Kaufman, Y. J., Stark, R., & Rundquist, D. (2002). Novel algorithms for remote estimation of vegetation fraction. *Remote Sensing of Environment*, 80(1), 76–87.
- Guyon, I., & Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3, 1157–1182.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *Hierarchical clustering. Data mining, inference and prediction* (2nd ed.), Springer Series in Statistics New York, NY, USA: Springer.
- Hillnhütter, C., & Mahlein, A. K. (2008). Early detection and localization of sugar beet diseases: new approaches. *Gesunde Pflanzen*, 60(4), 143–149.
- Huang, Y., Lan, Y., Thomson, S. J., Fang, A., Hoffmann, W. C., & Lacey, R. E. (2010). Development of soft computing and applications in agricultural and biological engineering. *Computers and Electronics in Agriculture*, 71(2), 107–127.
- Hughes, G. F. (1968). On the mean accuracy of statistical pattern recognizers. *IEEE Transactions on Information Theory*, 14(1), 55–63.
- Jebara, T. (2004). *Machine Learning: Discriminative and Generative*, Springer Series in Engineering and Computer Science Berlin, Germany: Springer.
- Jensen, J.R. (2007). Remote sensing of the environment: an earth resource perspective 2nd. In: D. Kaveney (Ed.). Upper Saddle River, NJ, USA: Prentice Hall.
- Jolliffe, I. T. (2002). *Principal component analysis*, Springer series in statistics New York, NY, USA: Springer.
- Karimi, Y., Prasher, O. S., Patel, R. M., & Kim, H. S. (2006). Application of support vector machine technology for weed and nitrogen stress detection in corn. *Computers and Electronics in Agriculture*, 51(1–2), 99–109.
- Kohavi, R., & John, H. G. (1997). Wrappers for feature subset selection. *Artificial Intelligence*, 97(1–2), 273–324.
- Lafferty, J., Pereira, F., McCallum, A. (2001). Conditional random fields: probabilistic models for segmenting and labelling sequence data. In: C. E. Brodley, A. P. Danyluk (Eds.), *Proceedings of the 18th International Conference on Machine Learning 2001 (ICML 2001)* (pp 282–289). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.
- Lamb, D. W., & Brown, R. B. (2001). PA-Precision agriculture: remote-sensing and mapping of weeds in crops. *Journal of Agricultural Engineering Research*, 78(2), 117–125.
- Larios, N., Soran, B., Shapiro, L. G., Martinez-Munoz, G., Lin, J., Dietterich, T. G. (2010). Haar random forest features and SVM spatial matching kernel for stonefly species identification. *Proceedings of IEEE International Conference on Pattern Recognition (ICPR)*, IEEE, (pp. 2624–2627). Istanbul, Turkey.
- Laudien, R., Bareth G., Doluschitz, R. (2003). Analysis of hyperspectral field data for detection of sugar beet diseases. In: Z. Harnos, M. Herdon, T. B. Wiwczarowski. (Eds.), *Proceedings of the 4th European Federation for Information Technology in Agriculture, Food and the Environment (EFITA) Conference* (pp. 375–381). Debrecen-Budapest, Hungary.
- Lee, W. S., Alchanatis, V., Yang, C., Hirafuji, M., Moshou, D., & Li, C. (2010). Sensing technologies for precision specialty crop production. *Computers and Electronics in Agriculture*, 74(1), 2–33.
- Liu, Z.-Y., Wu, H.-F., & Huang, J.-F. (2010). Application of neural networks to discriminate fungal infection levels in rice panicles using hyperspectral reflectance and principal components analysis. *Computer and Electronics in Agriculture*, 72(2), 99–106.
- Longchamps, L., Panneton, B., Samson, G., Leroux, G., & Thériault, R. (2010). Discrimination of corn, grasses and dicot weeds by their UV-induced fluorescence spectral signature. *Precision Agriculture*, 11, 181–197.
- Lopez-Granados, F., Pena-Barragan, J. M., Jurado-Exposita, M., Francisco-Fernandez, M., Cao, R., Alosno-Betanzos, A., et al. (2008). Multispectral classification of grass weeds and wheat (*Triticum durum*) using linear and nonparametric functional discriminant analysis and neural networks. *Weed Research*, 48(1), 28–37.

- Mahlein, A.-K., Oerke, E.-C., Steiner, U., & Dehne, H.-W. (2012a). Recent advances in sensing plant diseases for precision crop protection. *European Journal of Plant Pathology*, 133(1), 197–209.
- Mahlein, A.-K., Rumpf, T., Welke, P., Dehne, H.-W., Plümer, L., Steiner, U., et al. (2013). Development of spectral vegetation indices for detecting and identifying plant diseases. *Remote Sensing of Environment*, 128, 21–30.
- Mahlein, A.-K., Steiner, U., Hillnhütter, C., Dehne, H.-W., & Oerke, E.-C. (2012b). Hyperspectral imaging for small-scale analysis of symptoms caused by different sugar beet diseases. *Plant Methods*, 8(1), 3.
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge, UK: Cambridge University Press.
- Martínez, A. M., & Kak, A. C. (2001). Pca versus lda. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 23(2), 228–233.
- Mewes, T., Franke, J., & Menz, G. (2011). Spectral requirements on airborne hyperspectral remote sensing data for wheat disease detection. *Precision Agriculture*, 12(6), 795–812.
- Mirik, M., Michels, G. J. Jr, Kassymzhanova-Mirik, S., & Elliott, N. C. (2007). Reflectance characteristics of Russian wheat aphid (*Hemiptera: aphididae*) stress and abundance in winter wheat. *Computers and Electronics in Agriculture*, 57(2), 123–134.
- Mitchell, T. M. (1997). *Machine Learning*, McGraw-Hill Series in Computer Science New York, NY, USA: McGraw-Hill Science.
- Moshou, D., Bravo, C., Oberti, R., West, J., Bodria, L., McCartney, A., et al. (2005). Plant disease detection based on data fusion of hyper-spectral and multi-spectral fluorescence imaging using kohonen maps. *Real Time Imaging Journal-Special Issue on spectral Imaging II*, 11(2), 75–83.
- Moshou, D., Bravo, C., Oberti, R., West, J. S., Ramon, H., Vougioukas, S., et al. (2011). Intelligent multi-sensor system for the detection and treatment of fungal diseases in arable crops. *Biosystems Engineering*, 108(4), 311–321.
- Moshou, D., Bravo, C., West, J., Wahlen, S., McCartney, A., & Ramon, H. (2004). Automatic detection of “yellow rust” in wheat using reflectance measurements and neural networks. *Computers and Electronics in Agriculture*, 44(3), 173–188.
- Moshou, D., Ramon, H., & De Baerdemaeker, J. (2002). A weed species spectral detector based on neural networks. *Precision Agriculture*, 3(3), 209–223.
- Mucherino, A., Papajorgji, P., & Paradalos, M. P. (2009). A survey of data mining techniques applied to agriculture. *Operational Research*, 9(2), 121–140.
- Nieuwenhuizen, A. T., Tang, L., Hofstee, J. W., Müller, J., & Van Henten, E. J. (2007). Colour based detection of volunteer potatoes as weeds in sugar beet fields using machine vision. *Precision Agriculture*, 8(6), 267–278.
- Oerke, E.-C., & Dehne, H.-W. (2004). Safeguarding production—losses in major crops and the role of crop protection. *Crop Protection*, 23, 275–285.
- Oumar, Z., Mutanga, O., & Ismail, R. (2013). Predicting *Thaumastocoris peregrinus* damage using narrow band normalized indices and hyperspectral indices using field spectra resampled to the Hyperion sensor. *International Journal of Applied Earth Observation and Geoinformation*, 21, 113–121.
- Panneton, B., Guillaume, S., Roger, J.-M., & Samson, G. (2010). Improved discrimination between monocotyledonous and dicotyledonous plants for weed control based on the blue-green region of ultraviolet-induced fluorescence spectra. *Applied Spectroscopy*, 64(1), 30–36.
- Patterson, D. T. (1995). Weeds in a changing climate. *Weed Science*, 43(4), 685–701.
- Perez, A. J., Lopez, F., Benlloch, J. V., & Christensen, S. (2000). Colour and shape analysis techniques for weed detection in cereal crops. *Computers and Electronics in Agriculture*, 25(3), 197–212.
- Potamitis, I., Ganchev, T., Fakotakis, N. (2007). Automatic acoustic identification of crickets and cicadas. In: M. Al-Mualla (Ed.), *Proceedings of the International Symposium of Signal Processing and Its Applications (ISSPA)*, IEEE, (pp. 1–4).
- Prabhakar, M., Prasad, Y. G., Thirupathi, M., Sreedevi, G., Dharajothi, B., & Venkateswarlu, B. (2011). Use of ground based hyperspectral remote sensing for detection of stress in cotton caused by leafhopper (*Hemiptera: Cicadellidae*). *Computers and Electronics in Agriculture*, 79(2), 189–198.
- Pydipati, R., Burks, T. F., & Lee, W. S. (2005). Statistical and neural network classifiers for citrus disease detection using machine vision. *Transactions of the ASAE*, 48(5), 2007–2014.
- Rakotomamonjy, A. (2003). Variable selection using SVM based criteria. *The Journal of Machine Learning Research*, 3, 1357–1370.
- Robnik-Šikonja, M., & Kononenko, I. (2003). Theoretical and empirical analysis of relief and relieff. *Machine Learning*, 53(1), 23–69.
- Römer, C., Bürling, K., Hunsche, M., Rumpf, T., Noga, G., & Plümer, L. (2011). Robust fitting of fluorescence spectra for pre-symptomatic wheat leaf rust detection with support vector machines. *Computers and Electronics in Agriculture*, 79(2), 180–188.

- Rosenblatt, F. (1962). *Principles of neurodynamics: perceptrons and the theory of brain mechanisms*. Washington, DC, USA: Spartan.
- Rumpf, T., Mahlein, A. K., Steiner, U., Oerke, E. C., Dehne, H. W., & Plümer, L. (2010). Early detection and classification of plant diseases with support vector machines based on hyperspectral reflectance. *Computers and Electronics in Agriculture*, 74(1), 91–99.
- Rumpf, T., Römer, C., Weis, M., Sökefeld, M., Gerhards, R., & Plümer, L. (2012). Sequential support vector machine classification for small-grain weed species discrimination with special regard to *Cirsium arvense* and *Gallium aparine*. *Computers and Electronics in Agriculture*, 80, 89–96.
- Saeys, Y., Inza, I., & Larrañaga, P. (2007). A review of feature selection techniques in bioinformatics. *Bioinformatics*, 23(19), 2507–2517.
- Schölkopf, B., & Smola, A. J. (2002). *Learning with kernels: support vector machines, regularization, optimization, and beyond*. Cambridge, MA, USA: MIT University Press.
- Setti, B., Bencheikh, M., Henni, J., & Neema, C. (2009). Comparative aggressiveness of *Mycosphaerella pinodes* on peas from different regions in western Algeria. *Phytopathologia Mediterranea*, 48(2), 195–204.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(379–423), 623–656.
- Stafford, J. V. (2000). Implementing precision agriculture in the 21st Century. *Journal of Agricultural Engineering Research*, 76(3), 267–275.
- Suzuki, K. (2011). *Artificial neural networks—methodological advances and biomedical applications.*, InTech series of numerical analysis and scientific computing Rijeka, Croatia: InTech.
- Suzuki, Y., Okamoto, H., & Kataoka, T. (2008). Image segmentation between crop and weed using hyperspectral imaging for weed detection in soybean field. *Environmental Control Biology*, 46(3), 163–173.
- Tang, L., Tian, L., & Steward, B. L. (2000). Color image segmentation with genetic algorithm for in-field weed sensing. *Transactions of the ASAE*, 43(4), 1019–1027.
- Thorp, K. R., & Tian, L. F. (2004). A review on remote sensing of weeds in agriculture. *Precision Agriculture*, 5(5), 477–508.
- Timmermann, C., Gerhards, R., & Kühbauch, W. (2003). The economic impact of site-specific weed control. *Precision Agriculture*, 4(3), 249–260.
- Tu, J. V. (1996). Advantages and disadvantages of using artificial neural networks versus logistic regression for predicting medical outcomes. *Journal of Clinical Epidemiology*, 49(11), 1225–1231.
- Vinzi, V. E., Chin, W. W., Henseler, J., & Wang, H. (Eds.). (2010). *Handbook of partial least squares: concepts, methods and applications.*, Springer Handbooks of Computational Statistics Berlin, Germany: Springer.
- Waibel, A., Hanazawa, T., Hinton, G., Shikano, K., & Lang, K. J. (1989). Phoneme recognition using time-delay neural networks. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 37(3), 328–339.
- Wang, X., Zhang, M., Zhu, J., & Geng, S. (2008). Spectral prediction of *Phytophthora infestans* infection on tomatoes using artificial neural network (ANN). *International Journal of Remote Sensing*, 29(6), 1693–1706.
- Waske, B., van der Linden, S., Benediktsson, J. A., Rabe, A., & Hostert, P. (2010). Sensitivity of support vector machines to random feature selection in classification of hyperspectral data. *IEEE Transactions on Geoscience and Remote Sensing*, 48(7), 2880–2889.
- Wetterich, C.B., Kumar, R., Sankaran, S., Junior, J.B., Eshani, R., Marcassa, L.G. (2013). A comparative study on application of computer vision and fluorescence imaging spectroscopy for detection of Huanglongbing citrus disease in the USA and Brazil. *Journal of Spectroscopy*, Article ID 941738. 6 pages.
- Witten, I. A., & Frank, E. (2005). *Data mining: practical machine learning tools and techniques* (2nd ed.), The Morgan Kaufmann Series in Data Management Systems San Francisco, CA, USA: Morgan Kaufmann.
- Wu, D., Feng, L., Zhang, C., & He, Y. (2008). Early detection of *Botrytis cinerea* on eggplant leaves based on visible and near-infrared spectroscopy. *Transactions of the ASABE*, 51(3), 1113–1139.
- Wu, L., & Wen, Y. (2009). Weed/corn seedling recognition by support vector machine using texture features. *African Journal of Agricultural Research*, 4, 840–846.
- Yang, C.-C., Prasher, S. O., Landry, J.-A., & Ramaswamy, H. S. (2003). Development of a herbicide application map using artificial neural networks and fuzzy logic. *Agricultural Systems*, 76(2), 561–574.
- Zhang, M., & Meng, Q. (2011). Automatic citrus canker detection from leaf images captured in field. *Pattern Recognition Letters*, 32(15), 2036–2046.
- Zhao, Y., He, Y., & Xu, X. (2012). A novel algorithm for damage recognition on pest-infested oilseed rape leaves. *Computers and Electronics in Agriculture*, 89, 41–50.